
SkillClaw

Let Skills Evolve Collectively with Agentic Evolver

Collective Skill Evolution for Multi-User LLM Agent Ecosystems



arxiv.org/abs/2604.08377

Agent Skills Remain Static After Deployment

✗ Static Skill Paradigm (Current)

Skills manually installed from centralized hub

Solutions discovered during interaction
never persist beyond individual sessions

Same patterns of failure and recovery
repeatedly observed across users

Each user forced to rediscover solutions
independently

EXISTING APPROACHES FALL SHORT

Memory-based: Store trajectories but tied to
specific instances, hard to generalize

Skill-based: Compress into structured instructions
but treat library as static resource

✓ Evolving Skill Paradigm (SkillClaw)

Skills continuously evolve from
runtime interaction evidence

Cross-user trajectories aggregated
for complementary behavioral signals

Agentic evolver performs open-ended
reasoning over evidence

Updated skills verified and synced
back to all agents automatically

Key Insight

Knowledge accumulates at the system level,
not just within individual sessions.

Closed-Loop Pipeline: Multi-User Interaction to Shared Skill Evolution



How It Works

Independent agents interact with environments and produce structured session trajectories preserving full action-feedback causal chains.

Trajectories are aggregated across users, grouped by referenced skills, forming a shared evidence base for the agentic evolver.

The evolution process is driven by an autonomous agent performing open-ended reasoning over interaction evidence.

Agentic Evolver: Evidence → Attribution → Evolution

1 Evidence



Analyze trajectories to identify behavioral patterns

Structured session trajectories with full action-feedback causal chains are collected and grouped by referenced skills.

INPUT: Cross-user trajectories

2 Attribution



Diagnose root causes of failures and successes

Identify which skill behaviors led to each outcome.

Separate generalizable improvements from idiosyncratic fixes.

PROCESS: Root-cause diagnosis

3 Evolution



Propose skill mutations — refine existing or create new

Updated skills are verified before being merged into the shared repository.

Conservative verification prevents regressions.

OUTPUT: Verified skill updates



Key Distinction

Skill updates are produced through open-ended reasoning by an autonomous agent — not predefined update rules.

This enables context-aware mutations that adapt to observed failure patterns rather than applying generic fixes.

Cross-User Trajectory Aggregation Exposes Behavioral Boundaries

Why Multi-User Signals Matter

Different users exercise the same skill under diverse contexts, producing complementary views of that skill's behavioral boundary.

Key Insight

A single user rarely generates enough signal to separate a *generalizable improvement from an idiosyncratic fix*.

Reveals both conditions under which a skill works and those under which it breaks.

AGGREGATION PROCESS



Compatible Claw-Style Agent Systems

OpenClaw

CoPaw

IronClaw

PicoClaw

ZeroClaw

NanoClaw

NemoClaw

Evaluated on WildClawBench using Qwen3-Max as backbone, demonstrating substantial improvements across tasks.

Case Study: Multimodal Creative Pipeline — 1 of 6 Accepted

Conservative verification in action

DAY	CANDIDATE SKILL	SKILL FUNCTION	RESULT	NEXT-DAY ACTION
1	validate-tmp-workspace-inputs	Check /tmp_workspace inputs & environment setup	✔ Accept	Upgrade to new best pool
2	multimodal-input-validation-and-setup	Multimodal input validation & output env init	✘ Reject	Keep current best pool
3	multimodal-creative-task-pipeline	Multimodal creative pipeline (extract from PDF/video/image)	✘ Reject	Keep current best pool
4	multimodal-creative-task-pipeline (improved)	Added image classification, visual generation, validation	✘ Reject	Keep current best pool
5	pipeline (impr.) + validate-required-input-files	Creative pipeline + per-file fail-fast validation	✘ Reject	Keep current best pool
6	multimodal-creative-task-pipeline (candidate)	Extended PDF-to-poster / document-to-visual paths	✘ Reject	Not admitted to next cycle

Takeaway: Only 1 of 6 candidates accepted over 6 days — conservative verification prevents regressions and protects skill quality.

Acceptance rate: 16.7% | The system favors stability over novelty when evidence does not support improvement.

Case Study: Git Push Auth-Fallback — 3 of 6 Accepted

Iterative refinement then plateau

DAY	CANDIDATE SKILL	CHANGE SUMMARY	RESULT	NEXT-DAY ACTION
1	git-push-with-auth-fallback	Safe fallback instead of blocking on push failure	✓ Accept	Add to Safety best pool
2	git-push-with-auth-fallback	Unified patch/bundle filenames, reduced inconsistency	✓ Accept	Keep updated best pool
3	auth-fallback + git-clone-to-directory	Auth-alternative paths & secrets audit; fixed subshell pitfalls	✓ Accept	Keep current best pool
4	(none — same-pool retest)	Validator confirmed current pool as best	✓ Confirm	Continue same best pool
5	git-push-with-auth-fallback	Added "push hang treated as auth failure"; no improvement	✗ Reject	Keep current best pool
6	git-push-with-auth-fallback	Added identity config & filename consistency; no improvement	✗ Reject	Not admitted to next cycle

Takeaway: 3 of 6 days yielded accepted updates — successful iterative refinement before reaching a natural plateau.

Acceptance rate: 50% | Contrast with creative pipeline (16.7%): when evidence supports improvement, the system accepts updates.

Three Properties That Distinguish SkillClaw

1

Collective Evolution

Knowledge from individual interactions contributes to a shared, continuously improving skill ecosystem.

Cross-user trajectory aggregation provides complementary behavioral signals that a single user cannot generate alone.

Individual sessions → System-level knowledge accumulation

2

Fully Automatic

Skill evolution is driven by runtime interaction without manual curation or explicit user intervention.

No human-in-the-loop required — the system autonomously identifies improvement opportunities from usage patterns.

Zero manual curation → Continuous autonomous improvement

3

Agentic Evolution Paradigm

Skill updates are produced through open-ended reasoning rather than predefined update rules.

The agentic evolver's Evidence → Attribution → Evolution pipeline enables context-aware skill mutations.

Predefined rules → Open-ended agent reasoning over evidence



Implication: SkillClaw points toward a future where agent systems continuously improve through collective usage — every interaction becomes a potential source of system-wide enhancement.